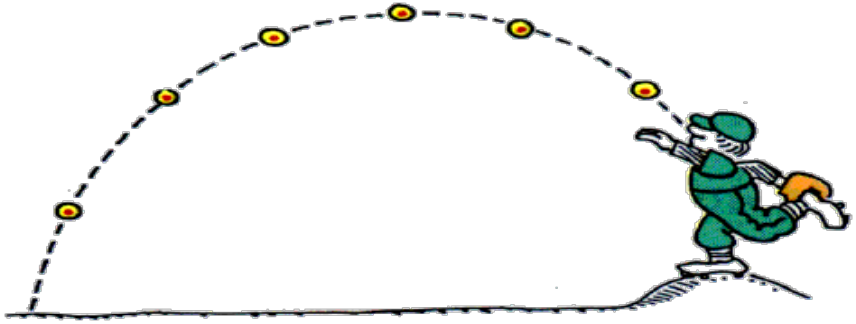
Chapter 9: Linear Momentum

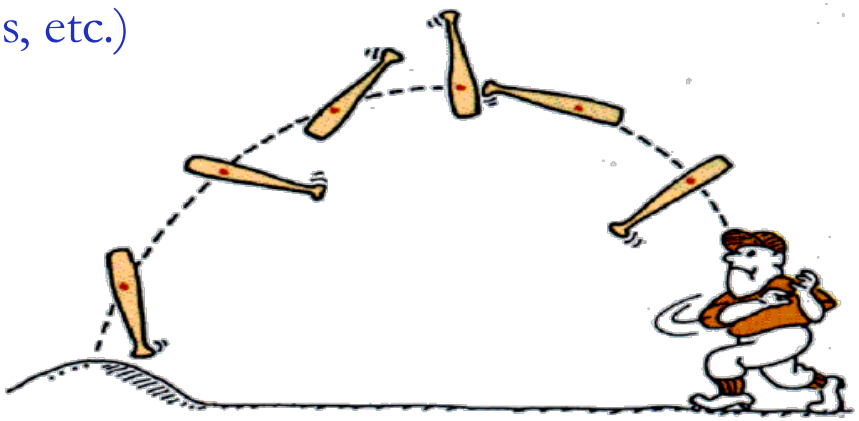
The following subjects will be discussed

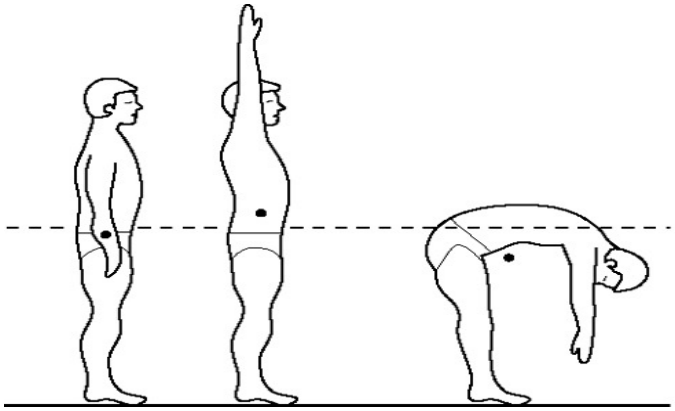
- Center of Mass.
- Linear Momentum.
- Linear Momentum Conservation Law.
- Linear Momentum and Kinetic Energy.
- Collisions.

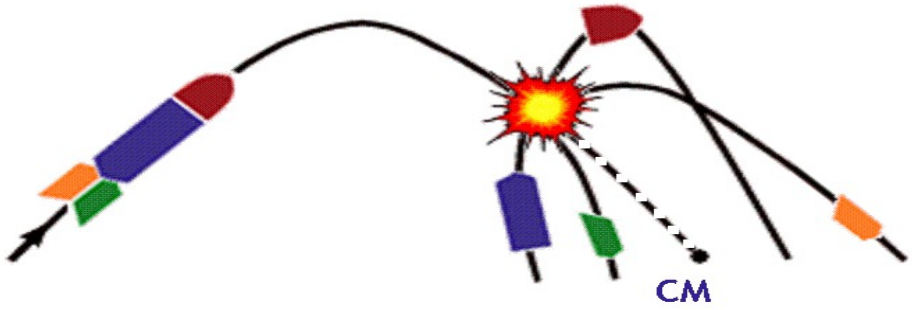
The Center of Mass

We define the center of mass (cm) for objects(s) that cannot be viewed as point particles (irregular shapes, groups of particles, etc.)









For irregular objects or a group of particles, the equations of motion (Newton's laws) are applied to the center of mass of the system.

Center of Mass Calculations

Consider two small objects arranged in 1D as shown below.

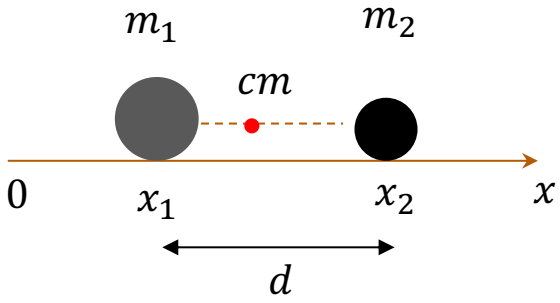
$$x_{cm} = \frac{x_1 m_1 + x_2 m_2}{m_1 + m_2}$$

$$x_{cm} = \frac{m_2}{m_1 + m_2} d$$

From 1

$$x_{cm} = \frac{m_1}{m_1 + m_2} d$$

From 2



$$x_{cm} = \frac{1}{M} \sum_{i=1}^{n} x_i \, m_i$$

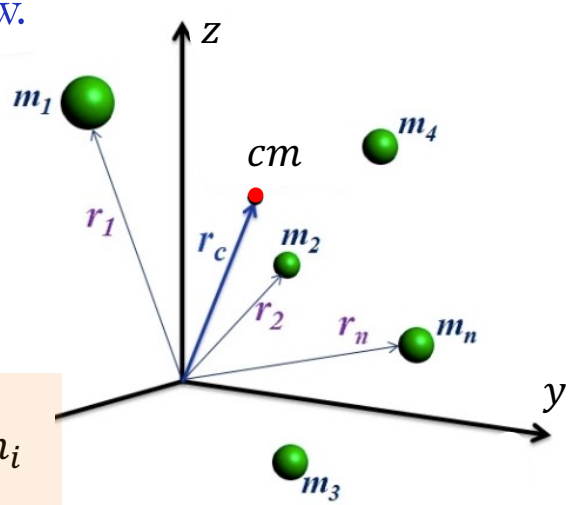
Consider many small objects arranged in 3D as shown below.

$$\vec{r}_{cm} = x_{cm}\,\vec{\imath} + y_{cm}\,\vec{\jmath} + z_{cm}\,\vec{k}$$

$$x_{cm} = \frac{x_1 m_1 + \cdots + x_n m_n}{m_1 + \cdots + m_n}$$

$$y_{cm} = \frac{y_1 m_1 + \cdots + y_n m_n}{m_1 + \cdots + m_n}$$

$$z_{cm} = \frac{z_1 m_1 + \cdots + z_n m_n}{m_1 + \cdots + m_n}$$



$$\vec{r}_{cm} = \frac{1}{M} \sum_{i=1}^{n} \vec{r}_i \, m_i$$

For a many-particle solid object

$$x_{cm} = \frac{1}{M} \int x \, dm$$

Center of Mass Dynamics

We can have a collective description of the motion of the multi-particle system by applying the equations of motion to the center of mass.

Newton's Law:

$$\vec{F}_{net} = M\,\vec{a}_{cm}$$

- $\vec{F}_{net}$: Net external forces
- M: Total mass
- $\vec{a}_{cm}$: Center of mass acceleration

$$\vec{F}_{net_x} = M\,\vec{a}_{cm_x} \qquad \vec{F}_{net_y} = M\,\vec{a}_{cm_y} \qquad \vec{F}_{net_z} = M\,\vec{a}_{cm_z}$$

We can write the previous summation as

$$\vec{r}_{cm} = \frac{1}{M}\sum_{i=1}^{n} \vec{r}_i\, m_i$$

$$M\,\vec{r}_{cm} = m_1\vec{r}_1 + m_2\vec{r}_2 + \cdots + m_n\vec{r}_n$$

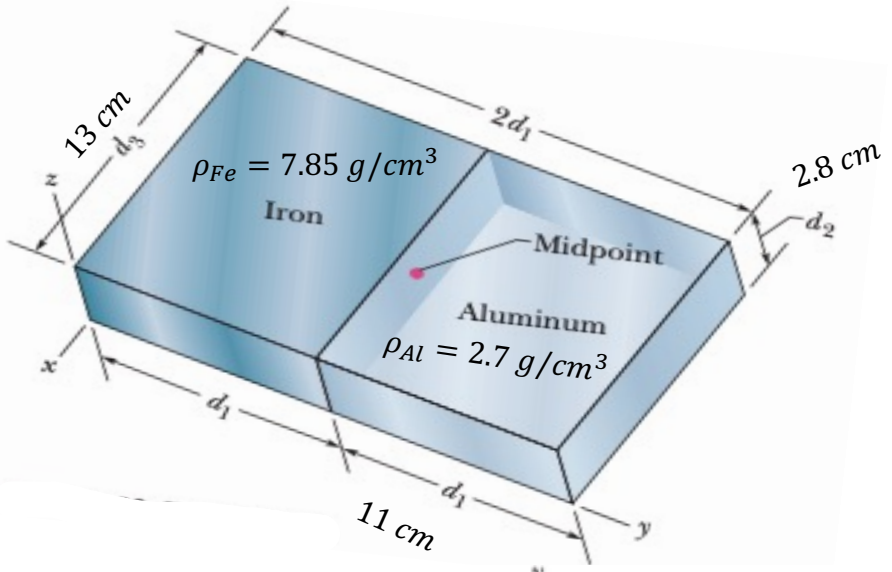
$$M\,\vec{v}_{cm} = m_1\vec{v}_1 + m_2\vec{v}_2 + \cdots + m_n\vec{v}_n$$

Linear momentum

$$M\,\vec{a}_{cm} = m_1\vec{a}_1 + m_2\vec{a}_2 + \cdots + m_n\vec{a}_n$$

Examples

1) Find the center of mass.



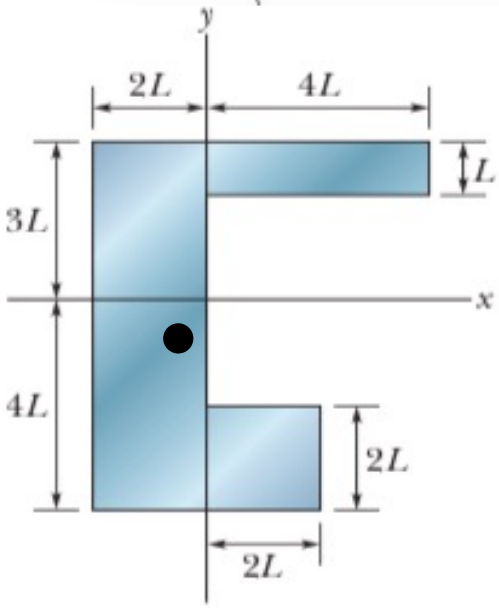
No change in the x direction $x_{cm} = \frac{d_3}{2}$

No change in the z direction $z_{cm} = \frac{d_2}{2}$

In the y direction $y_{cm} = \frac{d_1}{2} + \frac{\rho_{Fe}}{\rho_{Fe} + \rho_{Al}} d_1$

$(6.5\ cm, 8.3\ cm, 1.4\ cm)$

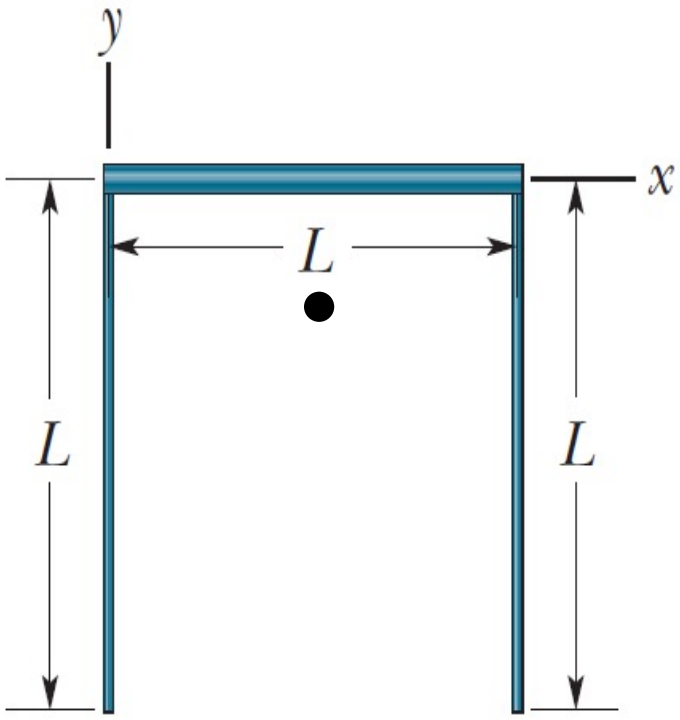
2) The structure shown in the figure is made of a uniform material. Find the x and y coordinates of the center of mass of the structure. Take L=5.0 m.

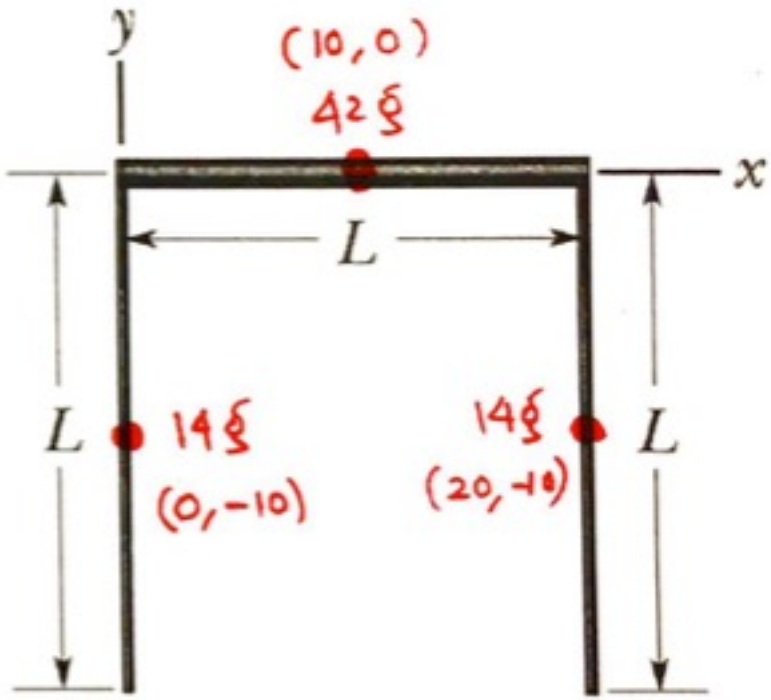


$(-0.45\ cm, -2\ cm)$

Examples

3) The structure shown in the figure is made of a uniform material. The vertical rods each have a mass of 14 g; the horizontal rod has a mass of 42 g. Find the x and y coordinates of the center of mass of the structure. Take L=20 cm.

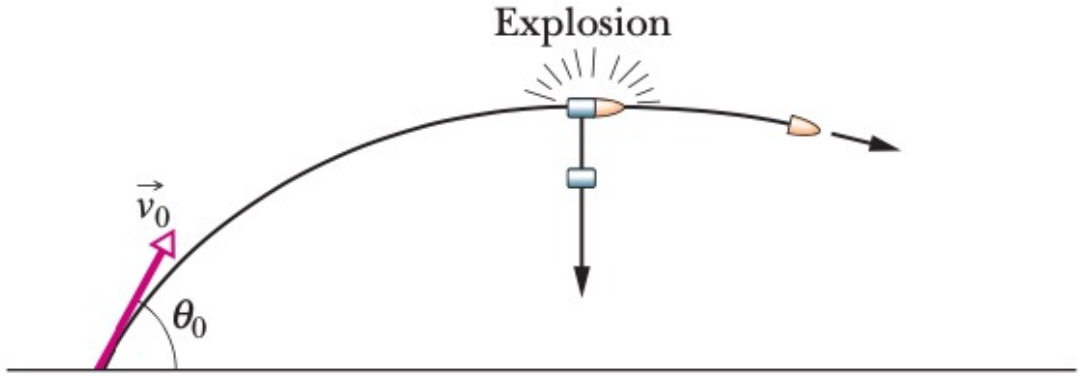




$(10\ cm, -4\ cm)$

Examples

4) The initial velocity of the object is 20 m/s at angle of 60°. At the top of the flight, the object breaks into two equal-mass parts. One part drops vertically at that point. Where does the other part fall?



Follow the center of mass.

$$Range = R = \frac{v_o^2}{g}\sin 2\theta$$

$$R = 35.3\ m$$

$$M\,\vec{r}_{cm} = m_1\vec{r}_1 + m_2\vec{r}_2 + \cdots + m_n\vec{r}_n$$

$$M\,x_{cm} = m_1x_1 + m_2x_2$$

$$M\,R = \left(\frac{M}{2}\right)\left(\frac{R}{2}\right) + \left(\frac{M}{2}\right)x_2$$

$$x_2 = 2R - \frac{R}{2} = 53\ m$$

Linear Momentum

We define the following quantity as the linear momentum of a moving object.

$$Linear\ Momentum = \vec{p} = m\,\vec{v}$$

Any moving particle has a kinetic energy and a linear momentum.

Take the time derivative of both sides: the rate of change of momentum.

net change in momentum

$$\frac{d\vec{p}}{dt} = m\frac{d\vec{v}}{dt} = m\,\vec{a} = \vec{F}$$

net external force

Meaning: External forces cause a change in object momentum.

Any change in object momentum points to the existence of external forces

If NO net external forces are applied to a particle, the momentum is conserved.

Linear Momentum

For a system of particles, the total momentum is given by,

$$\vec{P} = \vec{p}_1 + \vec{p}_2 + \cdots + \vec{p}_n$$

$$\vec{P} = m_1\vec{v}_1 + m_2\vec{v}_2 + \cdots + m_n\vec{v}_n$$

$$\vec{P} = M_{tot}\,\vec{v}_{cm}$$

Take the time derivative of both sides: the rate of change of momentum.

net change in system momentum

$$\frac{d\vec{P}}{dt} = M\frac{d\vec{v}_{cm}}{dt} = M\,\vec{a}_{cm} = \vec{F}$$

net external force

In the absence of net external forces, the system momentum cannot change.
It is conserved.

However, the individual momenta can change (internal forces).

Conservation of Linear Momentum

For an isolated, closed system with $\vec{F}_{net} = 0$,

$$\frac{d\vec{P}}{dt} = 0$$

$$\vec{P} = Constant$$

$$\vec{P}_{initial} = \vec{P}_{final}$$

Note that if $F_{net_x} = 0 \quad \rightarrow \quad P_x$ is conserved,

Impulse

When an object slams on a solid fixed target, there seem to be loss of momentum for a short period of time!

There is actually a force that compensates for the change in momentum, acting for a short time duration.

The force-time product is called impulse.

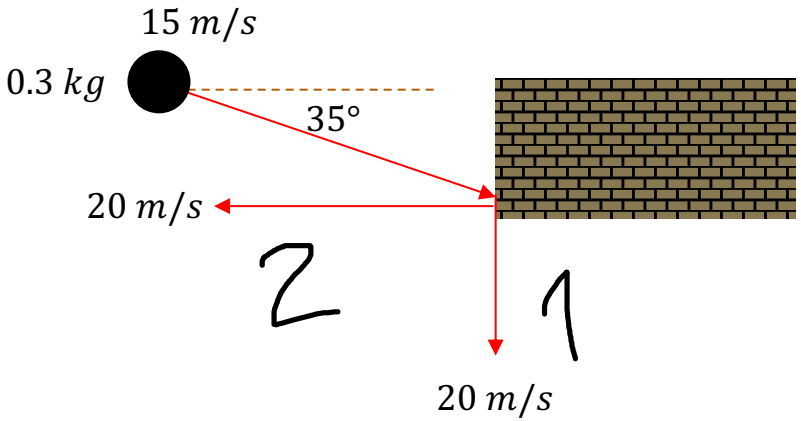


We can write $\frac{d\vec{p}}{dt} = \vec{F}$ as, $d\vec{p} = \vec{F}dt$ ⇨ $\int_{p_i}^{p_f} d\vec{p} = \int_{t_i}^{t_f} \vec{F}dt$

$p_f - p_i = [F(t_f) - F(t_i)](t_f - t_i)$ ⇨ $\Delta p = F_{ave}\ \Delta t = J$ Impulse (N.s)

Examples

1) Find the change in the ball momentum while in contact with the wall in both cases.



1

$$\Delta p_x = m\,\Delta v_x = m(0 - 15\cos 35) = -3.69\ \ kg\ m/s$$
$$\Delta p_y = m\,\Delta v_y = m(-20 + 15\sin 35) = -3.42\ \ kg\ m/s$$

$$|\Delta p| = 5\ kg\ m/s$$

2

$$\Delta p_x = m\,\Delta v_x = m(-20 - 15\cos 35) = -9.69\ \ kg\ m/s$$
$$\Delta p_y = m\,\Delta v_y = m(0 + 15\sin 35) = 2.58\ \ kg\ m/s$$

$$|\Delta p| = 10\ kg\ m/s$$

Examples

2) A 90-kg passenger experiences a free-fall elevator (36 m). Impact time is 5 ms with no rebound.



Impulse

$$|J| = \Delta p = m\,(v_f - v_i)$$

$$|J| = 90\left(0 - \sqrt{2(9.81)(36)}\right) = 2390\ N.s$$

$$F_{ave} = \frac{|J|}{\Delta t} = 478\ kN$$

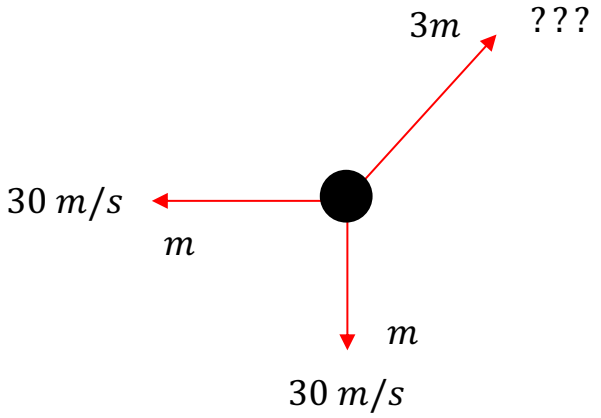
What if the passenger jumps up just before impact?

$$|J| = 90\left(0 - (\sqrt{2(9.81)(36)} - 7)\right) = 1760\ N.s$$

$$F_{ave} = \frac{|J|}{\Delta t} = 352.8\ kN$$

Examples

3) An object, initially at rest, explodes into three pieces, as shown. In what direction will the third piece go?



Conservation of momentum

$$\vec{P}_{initial} = \vec{P}_{final}$$

$$0 = \vec{p}_1 + \vec{p}_2 + \vec{p}_3$$

$$m_1\vec{v}_1 + m_2\vec{v}_2 + m_3\vec{v}_3 = 0$$

$$m\,\vec{v}_1 + m\,\vec{v}_2 + 3m\,\vec{v}_3 = 0$$

$$-30\,\vec{\imath} \;\; - 30\,\vec{\jmath} \;\; = -3\,\vec{v}_3$$

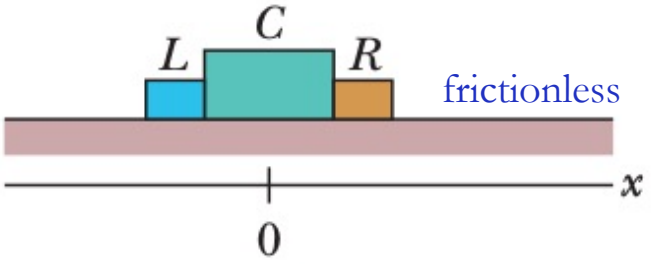
$$\vec{v}_3 = 10\,\vec{\imath} \;\; + 10\,\vec{\jmath} \;\; = 14.14\,\angle 45°\;\; m/s$$

Examples

4) The object (consisting of 3 parts, Center: 6 kg, Left: 2 kg, Right: 2 kg) is initially at rest.

1) At t=0, L fires and moves left with relative velocity 3 m/s.
2) At t=0.8 s, R fires and moves right with relative velocity 3 m/s.

Find velocity of C at t=2.8 s and its position.



Initially, $m_L v_L + m_C v_C + m_R v_R = 0 \rightarrow$

$$m_L v_L + m_{CR} v_{CR} = 0$$

1) $v_L = v_L' + v_{CR} \rightarrow \quad m_L(v_L' + v_{CR}) + m_{CR} v_{CR} = 0$

$$2(-3 + v_{CR}) + 8v_{CR} = 0 \rightarrow v_{CR} = 0.6\ m/s$$

2) $m_C v_C + m_R v_R = m_{CR} v_{CR} \qquad m_C v_C + m_R(v_R' + v_C) = m_{CR} v_{CR}$

$$6v_C + 2(3 + v_C) = 8(0.6) \rightarrow v_C = -0.15\ m/s$$

Find x_C at t=0.8 s, $(0.8)(0.6) = 0.48\ m$

Then it moved left until t=2.8 s, $(2.8 - 0.8)(-0.15) = -0.3\ m$

Final location

$x_C = 0.18\ m$

Q & A

Send me an email if
you have any
questions.